



GaugeHD Synthetic — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — GaugeHD Synthetic (MKM-GHD-001)

Tests run on **04 April 2026 at 18:43:46**.

Table 1: Test Results Summary — GaugeHD Synthetic (MKM-GHD-001)

Total Tests	62
Passed	62
Failed	0
Skipped	0
Duration	31.21s

Table 2: Detailed Test Results — GaugeHD Synthetic

Test Description	Acceptance Criteria	Result	Time
<code>_snap_to_river</code> skips zero-length segments without error.	<code>isinstance(lat, float);</code> <code>isinstance(lon, float)</code>	Pass	0.000s
<code>_create_synthetic_at_position</code> returns None when flanking gauge not in lookup.	<code>result is None</code>	Pass	0.004s
Lines 83-110: <code>generate_all_gauge_histories</code> processes all gauges.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.202s
Lines 105-107: error on one gauge -i logged, continues.	<code>isinstance(result, list)</code>	Pass	0.407s
All levels positive	<code>all(obs["level_meters"] > 0 for obs in result)</code>	Pass	0.007s
Correct length approximately	<code>728 <= len(result) <= 732</code>	Pass	0.007s
Dates are sequential	<code>(d_curr - d_prev).days == 1</code>	Pass	0.004s
Different seeds different output	<code>levels1 != levels2</code>	Pass	0.015s
Each obs has date and level	<code>"date" in obs; "level_meters" in obs</code> (+1 more)	Pass	0.002s
Gauge data without FloodGauge wrapper uses defaults.	<code>isinstance(result, list)</code>	Pass	0.002s
When historical high date is within range, that day gets historical_high level.	<code>match[0]["level_meters"] == 7.5</code>	Pass	0.008s
Invalid date format doesn't crash.	<code>isinstance(result, list)</code>	Pass	0.002s
Historical high date before start_date is ignored.	<code>isinstance(result, list)</code>	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Lines 176-179: Historical high injected via fallback when date not matched in loop.	<code>len(match) == 1;</code> <code>match[0]["level_meters"] == 8.0</code>	Pass	0.002s
Gauge data with no FloodStages uses sensible defaults.	<code>isinstance(result, list); len(result) > 0</code>	Pass	0.002s
Lines 97-100: Partially tidal gauge uses dampened tidal signal.	<code>len(result) > 700; all(lev > 0 for lev in levels)</code>	Pass	0.003s
Returns list	<code>isinstance(result, list)</code>	Pass	0.002s
Seed reproducible	<code>r1 == r2</code>	Pass	0.003s
Lines 92-95: Tidal gauge uses wider daily_std, tidal_amplitude > 0.	<code>len(result) > 700; max(levels) > min(levels)</code>	Pass	0.003s
Lines 147-149: Tidal component with spring-neap cycle.	<code>11 != 12</code>	Pass	0.003s
Lines 138-141: historical high injected after main loop.	<code>len(match) == 1;</code> <code>match[0]["level_meters"] == 9.99</code>	Pass	0.005s
When historical_high_dt is None (out of range), post-loop block skipped.	<code>all(o["level_meters"] != 8.0 for o in result)</code>	Pass	0.002s
When historical_high_date is None, both branches skipped.	<code>isinstance(result, list); len(result) > 0</code>	Pass	0.002s
generate() returns count=0 when GAUGE_POINTS has > 2 entries.	<code>result["count"] == 0</code>	Pass	0.001s
generate() returns count=0 when GAUGE_POINTS is None.	<code>result["count"] == 0</code>	Pass	0.001s
Lines 74-75: missing file raises FileNotFoundError.	—	Pass	0.000s
Lines 72-80: load_gauge_portfolio reads gauge.json.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.001s
_load_river_polyline returns None when cache file doesn't exist.	<code>result is None or isinstance(result, list)</code>	Pass	0.001s
_snap_to_river returns original coords when river is None.	<code>lat == 51.46; lon == -0.30</code>	Pass	0.000s
Synthetic gauges should not be placed at exact polyline endpoints.	<code>not (loc["GaugeLatitude"] == gp_lat and ...)</code>	Pass	0.403s
Lines 143-144: bad CSV -i error logged, continues.	<code>isinstance(result, list)</code>	Pass	0.002s
Lines 113-147: process_nrfa_directory scans CSV files.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.002s
Line 134: output_dir=None uses config.get_gaugehd_dir().	<code>isinstance(result, list)</code>	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Lines 131-147: no matching files -i empty list.	<code>result == []</code>	Pass	0.000s
Propertyhc has synth in nearest gauges	<code>found_synth</code>	Pass	0.005s
Synthetic gauge in propertyhc should have basis_bps computed.	<code>"basis_bps" in ng</code>	Pass	0.004s
Every property has one synthetic gauge	<code>not failures</code>	Pass	1.927s
Every property has three gauges total	<code>len(ngs) == 3</code>	Pass	0.149s
Every property has two real gauges	<code>not failures</code>	Pass	1.577s
Generator no longer needs property.json — uses random positions on river.	<code>result["count"] > 0</code>	Pass	0.028s
Generator should create up to the requested count of synthetic gauges.	<code>0 < result["count"] <= 3</code>	Pass	0.010s
Properties should have flood_events sourced from gauge storm responses.	<code>properties_with_events > 0</code>	Pass	1.587s
At least some properties should have flooded=True events.	<code>flooded_count > 0</code>	Pass	1.627s
Appends to gauge json	<code>len(data["flood_gauges"]) > 3;</code> <code>len(synth) > 0</code>	Pass	2.596s
Creates synthetic gauges	<code>result["count"] > 0;</code> <code>len(result["gauge_ids"]) == result["count"]</code>	Pass	2.562s
Synthetic gauges placed too close together should be deduplicated.	<code>result["count"] < 200;</code> <code>result["count"] > 0</code>	Pass	2.439s
Interpolated elevation	<code>3.0 <= elev <= 5.0</code>	Pass	2.491s
Interpolated flood stages	<code>3.5 <= stages["FloodAlert"] <= 4.5;</code> <code>4.5 <= stages["FloodWarning"] <= 5.5</code> (+1 more)	Pass	2.550s
No gauge json returns zero	<code>result["count"] == 0</code>	Pass	0.000s
Generator no longer needs property.json — generates random positions on river.	<code>result["count"] > 0</code>	Pass	2.549s
Requesting a small number of synthetic gauges should work.	<code>0 < result["count"] <= 5</code>	Pass	0.014s
Synthetic gauge has cdm structure	<code>"GaugeID" in fg["Header"];</code> <code>"CatchmentID" in fg["Header"]</code> (+8 more)	Pass	2.474s

Test Description	Acceptance Criteria	Result	Time
Synthetic gauge lat/lon should be on the polyline (very close to it).	<code>dist_m < 50.0; dist_m < 1.0</code>	Pass	2.691s
A pre-existing SYNTH- gauge in gauge.json is skipped in polyline matching.	<code>result["count"] >= 0</code>	Pass	2.474s
Gauge json has synth gauges	<code>len(synth) > 0</code>	Pass	0.002s
Synth gauge has required cdm fields	<code>fg["Header"]["GaugeID"].startswith("SYNTH")</code> <code>fg["Location"]["GaugeLatitude"] != 0</code> (+7 more)	Pass	0.002s
Gaugehc has synth curves	<code>len(synth) > 0</code>	Pass	0.013s
Synth has term structure	<code>len(ts) > 0</code>	Pass	0.011s
Synth hazard rates positive	<code>hc.get("annual_hazard_rate_alert", 0) > 0</code>	Pass	0.011s
Synth gaugets files exist	<code>len(synth_files) > 0</code>	Pass	0.001s
At least some storms should exceed alert at synthetic gauges.	<code>len(exceeded) > 0</code>	Pass	0.149s
Synth gaugets have storm responses	<code>len(resps) > 0</code>	Pass	0.171s

1.1 Test Document History

Date	Version	Description	Author
27-Mar-2026	1.0	Initial test suite (80 tests)	D. Kelly
03-Apr-2026	1.1	18 test(s) removed (total: 62)	D. Kelly
04-Apr-2026	1.2	5 test(s) added; 5 test(s) removed (total: 62)	D. Kelly